

ASSIGNMENT 5 (DUE DATE 27TH FEB)

- (1) Compute $A = H^*(S^n; \mathbb{Q})$ as a $\mathbb{Q}$ -algebra for $n > 1$ and find the minimal model of $(A, d = 0)$ for both even and odd case.
- (2) Compute $A = H^*(S^2 \vee S^2; \mathbb{Q})$ as a $\mathbb{Q}$ -algebra and find minimal model of $(A, d = 0)$. Extend the result for $S^n \vee S^m$ for $n, m \geq 2$.
- (3)** Compute $A = H^*(K(V, n); \mathbb{Q})$ as a $\mathbb{Q}$ -algebra, where V is a finite dimensional graded vector space over $\mathbb{Q}$. Compute the minimal model for A .
- (4)** Compute $A = H^*((S^2 \vee S^2) \times S^3; \mathbb{Q})$ as a $\mathbb{Q}$ -algebra and find its minimal model (M, d) . Show that exists (M', d') another minimal algebra which is not quasi-isomorphic to A but its cohomology agrees with A .